PwC UK Office Attendance Predictor

Project Proposal

Name: Emmanuel Phillips
Student ID: 230355585
Module: IOT6002U - Assessment 001
University email: ec23279@qmul.ac.uk
Work email: emmanuel.j.phillips@pwc.com

26 October 2026

Contents

1. Problem Definition 3
2. Project Aim and Objectives 4
3. Research Plan 5
4. Identification of Technologies, Tools and/or Datasets 6
5. Project Plan and Milestones 7
   5.1. Key dates 7
   5.2. Project build schedule (week 1 starts 2 Nov 2026, 25 weeks to submission) 7
6. Risk Register 8
7. References 9
8. Appendices 10
   8.1. Appendix A: Generative AI declaration 10
   8.2. Appendix B: Planned KSB mapping 10

1. Problem Definition

PwC UK operates a hybrid working model, so daily office attendance varies by site and weekday. Facilities Management (FM) uses attendance to plan catering and cleaning, and Real Estate to allocate space. Without forward visibility, FM risks food waste, mis-sized cleaning shifts and shortfalls on busy days.

Events such as bank holidays, rail strikes and internal events also shift attendance. These are logged in a manual form, but the log is historical and not used to forecast, even though many events are known in advance. Some public events (bank holidays, weather warnings) can be collected automatically, strikes can be compiled from published announcements, and internal events will remain manual.

The Real Estate Data Analytics (REDA) team's cloud data platform (Databricks) holds daily headcount per office. Forecasting relies on the built-in forecast in Power BI, a reporting tool, which has not been checked against past attendance or simple baselines, so its accuracy is unknown. Any new model must beat assuming attendance equals the same weekday last week, a standard benchmark (Makridakis, Spiliotis and Assimakopoulos, 2020).

2. Project Aim and Objectives

Aim: Develop and evaluate a 1-14 day office attendance forecasting pipeline that outperforms seasonal naive and Power BI baselines, collects public disruption data automatically and provides calibrated prediction intervals.

Objectives:

1. Define KPIs and simulation assumptions, and generate synthetic data mirroring REDA table structures.
2. Automate collection of bank holidays and weather warnings, compile announced strike dates, and store them in a staged (medallion) data architecture.
3. Benchmark seasonal naive and Power BI baselines.
4. Compare SARIMAX, LightGBM and XGBoost per office using rolling-origin validation (repeated forward forecasting; Cerqueira, Torgo and Mozetic, 2020) and average error measures (MAE, MAPE).
5. Produce prediction intervals and report their coverage.
6. Identify influential events using feature importance.
7. Deliver a dashboard, evaluate results and recommend next steps.

Future value: a validated forecast could support wider space optimisation decisions.

3. Research Plan

Prior work will be identified through IEEE Xplore, the ACM Digital Library, ScienceDirect, Google Scholar and arXiv, plus Databricks and MLflow documentation. Searches are mapped to project objectives (Table 1).

Peer-reviewed work from 2018 onwards is prioritised for methods, and from 2020 for hybrid working. Sources are selected for transparent data and relevance to small daily series. Occupancy estimation research is largely sensor-based (Chen, Jiang and Xie, 2018), so findings will guide method choices rather than set expected accuracy.

AI may support keyword generation and abstract screening; all sources are read and verified directly. References are managed using MyBib. Findings will inform model selection, validation, feature design and interval calibration.

Table 1: Literature search plan by project objective

Objective	Search terms	Sources	Purpose
Simulation	"synthetic time series generation" "hybrid working attendance patterns" "weekday effects office attendance"	Google Scholar, arXiv	Ground simulation assumptions and reduce the risk of unrealistic data
Platform	"medallion architecture" "MLflow experiment tracking"	Databricks and MLflow documentation	Inform pipeline design and experiment tracking
Baselines	"seasonal naive forecasting" "forecast benchmark" "M-competition"	IEEE Xplore, Google Scholar	Confirm baseline choice and standard evaluation metrics
Models	"SARIMAX exogenous variables" "LightGBM time series" "XGBoost forecasting" "rolling-origin evaluation"	ScienceDirect, arXiv	Justify model selection and validation strategy
Events	"calendar effects forecasting" "holiday effect forecasting" "special events demand forecasting" "feature importance time series"	ACM DL, IEEE Xplore	Decide which event features to include and how to encode them
Intervals	"conformal prediction time series" "prediction interval calibration" "quantile regression forecasting"	arXiv, Google Scholar	Choose an interval method suitable for small daily series
Context	"hybrid work office occupancy" "workplace utilisation post-2020"	Google Scholar, grey literature	Ground assumptions about attendance patterns

4. Identification of Technologies, Tools and/or Datasets

Platform: Databricks with Delta tables and MLflow (experiment tracking), so the pipeline mirrors production, with Unity Catalog providing governance. Alternatives: local Python with DuckDB (cheaper, ungoverned) and Azure ML (heavier for a small problem).

Models: Chosen for interpretability (SARIMAX; Hyndman and Athanasopoulos, 2021) and flexibility with event features (LightGBM, Ke et al., 2017; XGBoost, Chen and Guestrin, 2016), as compared in Table 2.

Data: Synthetic data is generated in Python (Faker plus custom simulation), mirroring the REDA attendance, office and date tables, with event effects from documented assumptions. Bank holidays, Met Office warnings and TfL disruption come from public feeds, and strike dates are compiled manually from union and operator announcements, as no structured feed exists. No real PwC attendance data is used.

Intervals and dashboard: Conformal prediction or quantile regression, chosen during research for small daily series; Power BI for the dashboard, with Databricks dashboards as an alternative.

Table 2: Candidate forecasting models, strengths, constraints and decision

Model	Strengths	Constraints	Decision
SARIMAX	Captures weekly patterns and event effects; easy to explain	Assumes simple relationships between events and attendance	Included
LightGBM	Quickly learns interacting effects (e.g. a strike on a Friday)	Harder to explain; can overfit short series	Included
XGBoost	Similar to LightGBM; widely used and documented	Harder to explain	Included
ARIMA	Simple classical method	Cannot use event data; SARIMAX extends it with seasonality and external variables	Excluded
Prophet	Handles holidays and seasonality automatically	Overlaps the included models; adds scope	Excluded
LSTM (neural network)	Learns long, complex patterns	Needs large data and is hard to explain; daily office series are small	Excluded

5. Project Plan and Milestones

5.1. Key dates

Table 3: Key project milestones and status

Status	Milestone	Date
Met	Project options shortlisted	10 Oct 2026
Planned	Discuss options with manager and lecturer	25 Oct 2026
Planned	Finalise choice of project	25 Oct 2026
Planned	Proposal submitted	30 Oct 2026
Planned	Project build starts (assumed)	2 Nov 2026
Planned	University exam period (expected)	6 to 20 Jan 2027
Planned	Return to PwC (expected)	First week of Feb 2027
Planned	Learning journals and portfolio (likely)	22 Feb 2027
Planned	Gateway (expected)	2nd to 3rd week of Mar 2027
Planned	Project report submission	23 Apr 2027
Planned	EPA sessions (online)	17 May to 4 Jun 2027

5.2. Project build schedule (week 1 starts 2 Nov 2026, 25 weeks to submission)

Table 4: Project build schedule by week from 2 Nov 2026 (week 25 ends at submission on 23 Apr 2027)

Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
KPIs, horizon and simulation assumptions	■	■																							
Synthetic data generator		■	■	■																					
Ingestion pipeline (bronze to gold)			■	■	■	■	■																		
Baselines (seasonal naive, Power BI)						■	■																		
Christmas and exams (no planned work)								■	■	■	■	■													
Model benchmarking in MLflow													■	■	■	■									
Check simulation assumptions with FM (at PwC)														■	■										
Intervals and feature importance																	■	■							
Dashboard																			■	■					
Dashboard feedback from FM and REDA (at PwC)																				■	■				
Evaluation and recommendations																				■	■				
Report drafting and finalising																		■	■	■	■	■	■	■	■

Dependencies: modelling depends on the synthetic data generator and ingestion pipeline. The dashboard depends on the chosen model and validated intervals. The build does not need the workplace, so it starts in November, with stakeholder checks (green) planned for when I am back at PwC. The report is drafted alongside evaluation and finalised after the gateway. If the schedule slips, the dashboard is dropped first.

6. Risk Register

Table 5: Risk register with likelihood, severity and mitigations

ID	Risk	Likelihood	Severity	Mitigation
R1	Synthetic data too simple, so models only recover the generating rules	High	High	Add noise, regime changes and unmodelled events, hold out event types from training, document all generator assumptions in an appendix, state that results show method performance rather than real-world accuracy
R2	Real PwC values or identifiers included by accident	Low	High	Generate from schema only, copy no real values, review all outputs before sharing
R3	Findings do not transfer to real data	Medium	Medium	Frame as a method evaluation, build the pipeline so real data can be swapped in later, base parameters on qualitative patterns described by FM
R4	Public feeds unreliable or change format	Medium	Medium	Store raw responses in bronze, validate on load, fall back to manual flag
R5	Models fail to beat baselines	Medium	Medium	Report as a valid finding, recommend baseline-plus improvements
R6	Scope creep	Medium	Medium	Fixed objectives, natural-language AI query tools (MCP, Genie) out of scope
R7	Confidentiality limits on describing PwC systems	Medium	Medium	Agree restrictions with manager on Power BI setup, office counts and schemas, use generic names
R8	Power BI baseline run on synthetic data is only a proxy for the live forecast	Medium	Low	Document the settings used, state the limitation in the evaluation
R9	Compute cost overrun	Low	Low	Small data, job clusters with auto-termination
R10	Forecast seen as employee monitoring, or reused for individual tracking or enforcing attendance	Medium	High	Use office-level daily totals only, with no individual or team data and a minimum group size; agree intended use with the line manager; exclude performance management use; review wording with FM and HR before any rollout
R11	Decisions based on the forecast disadvantage groups (e.g. carers, disabled or part-time staff) or give FM staff unpredictable shifts	Medium	High	Keep the forecast advisory with human sign-off; give minimum notice for shift changes; protect accessible and fixed desk provision; consult EDI and employee representatives before any production use

Ethics and compliance: The project uses synthetic data generated from table schemas only, with no real PwC values and no human participants. The module organiser does not foresee a need for university ethics review provided the work is not rolled out to production. Attendance forecasting still raises employee trust and EDI questions (R10, R11), so a data protection and EDI impact assessment would precede any rollout. Confidentiality limits on describing PwC systems will be agreed with the line manager.

7. References

Cerqueira, V., Torgo, L. and Mozetič, I. (2020) 'Evaluating time series forecasting models: an empirical study on performance estimation methods', *Machine Learning*, 109, pp. 1997–2028. Available at: https://doi.org/10.1007/s10994-020-05910-7.

Chen, T. and Guestrin, C. (2016) 'XGBoost: a scalable tree boosting system', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. New York: ACM, pp. 785–794. Available at: https://doi.org/10.1145/2939672.2939785.

Chen, Z., Jiang, C. and Xie, L. (2018) 'Building occupancy estimation and detection: a review', *Energy and Buildings*, 169, pp. 260–270. Available at: https://doi.org/10.1016/j.enbuild.2018.03.084.

Hyndman, R.J. and Athanasopoulos, G. (2021) *Forecasting: principles and practice*. 3rd edn. Melbourne: OTexts. Available at: https://otexts.com/fpp3/ (Accessed: 19 October 2026).

Ke, G. et al. (2017) 'LightGBM: a highly efficient gradient boosting decision tree', in *Advances in Neural Information Processing Systems 30*. Red Hook, NY: Curran Associates, pp. 3146–3154.

Makridakis, S., Spiliotis, E. and Assimakopoulos, V. (2020) 'The M4 Competition: 100,000 time series and 61 forecasting methods', *International Journal of Forecasting*, 36(1), pp. 54–74.

8. Appendices

8.1. Appendix A: Generative AI declaration

Generative AI was used in order to help explore potential risks, as well as non-trivial formatting. All AI output was read and verified before being inputted into my project proposal. I take full responsibility for the final content of this document.

8.2. Appendix B: Planned KSB mapping

Table 6: Planned KSB mapping

KSB	Planned evidence
S50	Dashboard and communication of results
S52, S53	Model benchmarking, rolling-origin validation and stability testing
S54, K56	Automated ingestion of public data sources
S55, K54	Analysis of simulated attendance across offices
S2, K3	Risk register and mitigations
S5, S6, K15	Project plan, schedule and time/cost estimation
K18, S14	Research plan and evaluation of forecasting methods
B3	Synthetic data approach, no real PwC data used